

Research Proposal — Structured Extraction & Review

Enhancing the competency of using problem solving strategy in mathematics among senior secondary teachers in Sri Lanka

Source: *Research proposal _ Navarajah Prasanthan_Signed.pdf* (14 pages)

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1 Proposal at a glance

Title	Enhancing the competency of using problem solving strategy in mathematics among senior secondary teachers in Sri Lanka
Candidate	Navarajah Prashanthan (P.hd Student, as printed)
Index no.	2023/Phd/PT/003
Institution	Faculty of Education, University of Colombo
Supervisor	Dr. D.V. K.P. Seneviratne — Head of the Department, Department of Science and Technology Education, Faculty of Education, University of Colombo, Colombo.
Supervisor sign-off	Signed, dated 23.01.2025 (recommendation text left blank)
Field	Mathematics education / teacher professional development
Population	Senior secondary (G.C.E. O/L, grades 10–11) mathematics teachers, Sri Lanka (confirmed by the candidate)
Design	Multi-phase mixed methods: quantitative survey → qualitative exploration → experimental/action intervention → teacher education module
Planned period	Jan 2025 – May 2026 (17 months)
Key deliverable	A teacher education module on effective use of problem-solving strategies in senior secondary mathematics

2 Document structure

Section	Content	PDF pp.
Front matter	Cover, topic page with index no., supervisor recommendation & signature	1–3
1 Introduction	1.1 Background of the study; 1.2 Research Problem	4–5
2 Literature Review	Four review themes + summary table of 5 studies	5–6
3 Research Objectives	Six objectives (I–VI), highlighted in source	7
4 Methodology	Design diagram; 4.1 Research Design (Phases 1–3, Participants); 4.2 Data Collection; 4.3 Data Analysis	8–11
5 Contributions	Potential contributions and implications	11
6 Timeline	Gantt chart, Jan 2025 – May 2026	12
7 Conclusion	One paragraph	12–13
8 References	21 entries	13–14

3 Research problem and rationale

3.1 Problem statement (condensed)

Senior secondary mathematics teachers in Sri Lanka possess subject knowledge but struggle to integrate problem-solving techniques into their teaching. Consequences named: low student

engagement, poor mathematics performance, weakened critical thinking. The study will *investigate and enhance* teacher competency in using problem-solving strategies.

3.2 Evidence claims made in the Introduction

Claim	Cited source	In reference list?
Room for improvement in teachers' mathematics knowledge	ICMI, 2016	No
Teachers' content knowledge insufficient; underprepared to help students internalise concepts	Siregar, 2017	Yes
Appointed mathematics teachers fail to upgrade their knowledge	National Education Commission, 2018	No
Performance of teachers in mathematics is poor	Department of Examination, 2019	Yes (DOE 2019 — an O/L examiners' report)

3.3 Key constructs

- **Problem-solving** — framed as a fundamental part of mathematical learning: exploring, conjecturing, reasoning.
- **Teacher competency** — knowledge + pedagogical skill to implement problem-solving strategies in class.
- **Cognitive abilities** of teachers regarding problem-solving strategies (Objective I).
- **Attitudes and beliefs** — introduced via the theoretical framework and dominant in the methodology text.

4 Literature review components

4.1 Planned review themes

Theoretical Framework	Theories on attitudes and beliefs: Theory of Planned Behavior; Self-Efficacy Theory
Previous Studies	Impact of teachers' attitudes and beliefs on instructional practice in mathematics
Problem-Solving Strategies	Effective strategies and their importance in secondary education
Contextual Factors in Sri Lanka	Cultural, societal and institutional influences on teaching practice

4.2 Studies tabulated

Author	Topic	Finding	Link to current study
NIE, 2019	Teacher' mathematics knowledge	Many teachers are without required knowledge and trainings to teach mathematics	It is essential to ensure all mathematics teachers are subject-competent and have good pedagogical knowledge to use

Author	Topic	Finding	Link to current study
			problem solving strategies to teach mathematics effectively.
DOE, 2018	Teachers' pedagogical knowledge	Many teachers fail to use proper pedagogy while teaching mathematics.	<i>(shared with row above)</i>
Vygotsky, 1978	Collaborative Learning	Studies have shown that collaborative approaches, where students work in groups to solve problems, enhance problem-solving skills. Vygotsky's Social Development Theory emphasizes the role of social interaction in cognitive development, supporting this finding	Design and implement an intervention to enhance teachers' competency in applying problem-solving strategies in mathematics
Lesh et al., 2019	Problem-Solving Strategies in Mathematics	An overview of effective problem-solving strategies and their importance in secondary education.	Empower mathematics problem solving competency in Sri Lanka

5 Objectives and traceability

No.	Objective	Addressed in
I	Identify the current level of cognitive abilities of secondary mathematics teachers regarding problem-solving strategies.	Phase 1, Phase 2
II	Explore the level of use of problem solving strategies in mathematics by senior secondary teachers.	Phase 1, Phase 2
III	Find out areas for improvement in problem-solving strategy among senior secondary teachers.	Phase 2
IV	Design and implement an intervention to enhance teachers' competency in applying problem-solving strategies in mathematics.	Phase 3
V	Assess the progress of use of problem-solving strategies in mathematics after intervention.	Phase 3
VI	Propose a teacher education module on effective use of problem solving strategies in mathematics in senior secondary education.	Phase 3

Phase 1 → Objectives I–II and Phase 2 → Objectives I–III are stated explicitly in the design diagram. The Phase 3 mapping (IV–VI) is inferred: Phase 3 “assesses progress after intervention” and is followed by the “Designing the Module” box; the source does not number these links.

6 Methodology — normalised by phase

6.1 Phase 1: Quantitative Phase

Method	Cross-sectional survey
Objectives	I, II
Sample	45 mathematics teachers from 5 education zones
Sampling	Stratified sampling
Data collection	Survey questionnaire (online or in person)
Analysis	• Factor analysis – factor loadings • Discriminant function analysis • Regression analysis

6.2 Phase 2: Qualitative Phase

Method	Exploratory study
Objectives	I, II, III
Sample	45 mathematics teachers from 5 education zones
Sampling	Subset of Phase 1 survey sample
Data collection	Individual in-depth semi-structured interviews with documents and observation
Analysis	• Coding and thematic analysis • Observation reports • Similar and different themes development • QSR N6 qualitative software

6.3 Phase 3: Quantitative & Qualitative mixed design using experimental design

Method	Intervention (experimental / action component)
Objectives	IV, V, VI
Sample	Convenient sample
Sampling	Convenience sampling
Data collection	Observation, reflection and documentation
Analysis	• Assess the progress of use of problem-solving strategies in mathematics after intervention • Feedback from teachers and stakeholders

6.4 Phase narrative (Section 4.1)

Step	Phase 1 — Survey	Phase 2 — Interviews	Phase 3 — Intervention
Design	Questionnaire on attitudes to mathematics & use of problem-solving strategies; perceptions,	Semi-structured interview protocol, open-ended questions with classroom examples	Experimental / action component; teachers co-design self-learning interventions

Step	Phase 1 — Survey	Phase 2 — Interviews	Phase 3 — Intervention
	effectiveness, barriers, facilitators		
Sampling	Purposive; “sufficient for statistical power”	Subset of survey sample, diverse in attitudes and contexts	Convenience sample
Collection	Online or in person	Individual or group interviews, recorded, with notes	Observation, reflection, documentation
Analysis / evaluation	Descriptive statistics; differences by experience, background, institution	Thematic analysis, compared with quantitative results	Effectiveness evaluation; feedback from teachers & stakeholders

6.5 Consolidated data plan (Sections 4.2–4.3)

Instrument	Purpose	Size
Structured questionnaire	Attitudes, beliefs, reported practices	Sample of 150 (convenience)
Semi-structured interviews	Experiences and challenges	30 teachers
Classroom observations	Actual implementation; triangulation	20 classes
Statistics	Correlation, regression (+ factor, discriminant analysis in diagram)	—
Thematic analysis	Interview transcripts & observation notes (QSR N6)	—

7 Timeline (calendar form)

Activity	Start	End	Months
Reading literature reviews	Jan 2025	Mar 2026	15
Writing 1 st and 2 nd Chapters	Jul 2025	Aug 2025	2
Developing data collection tools	Aug 2025	Oct 2025	3
Data collection	Sep 2025	Nov 2025	3
Data processing	Nov 2025	Feb 2026	4
Data analysis	Jan 2026	Mar 2026	3
Report writing	Jan 2026	May 2026	5
Thesis submission	Apr 2026	May 2026	2

8 Consistency review

Observations below compare the proposal against itself. They are offered as revision points, not as judgements on the study’s merit.

8.1 Sample size and sampling

Location	Sample	Sampling
Diagram, Phase 1	45 teachers, 5 education zones	Stratified
Diagram, Phase 2	45 teachers, 5 education zones	—
Diagram, Phase 3	—	Convenient
§4.1 Phase 1.2	“Sufficient number ... statistical power”	Purposive
§4.1 Phase 2.2	Subset of survey sample	—
§4.1 Participants	150 teachers, urban / suburban / rural	Convenient
§4.2	30 interviewees; 20 observed classes	—

- Four sample figures (45, 150, 30, 20) and three sampling schemes (stratified, purposive, convenience) are used for overlapping groups. One sampling frame per phase would fix this.
- Phase 2 has the same N as Phase 1 (45) yet is described as a subset; §4.2 gives 30.
- Factor analysis and discriminant function analysis with $N = 45$ are likely underpowered; the planned N and analyses should be aligned.

8.2 Construct alignment

- Title, problem and objectives focus on **competency / cognitive ability in problem-solving strategies**; the theoretical framework and much of §4.1 focus on **attitudes and beliefs towards mathematics** (Theory of Planned Behavior, Self-Efficacy). The link between the two is not stated.
- §4.1 Phase 2.3 refers to “self-learning ... G.C.E advanced level classes” and Phase 3 to “self-learning interventions”, which do not appear elsewhere and conflict with the O/L target population — this looks carried over from a different study.
- Population is “senior secondary” in the title but “secondary school mathematics teachers” in Participants. The candidate has confirmed the target is G.C.E. O/L (grades 10–11) teachers only; Participants should say so. DOE 2019 (an O/L examiners’ report) fits this target.
- §4.1 refers to “the research questions”, but no research questions are listed.

8.3 Design completeness

- Phase 3 is called experimental, but there is no control or comparison group, pre/post instrument, intervention length or content.
- Objective VI (the module) has no method beyond the “Designing the Module” box.
- No instrument development, validation or reliability testing (e.g. pilot, Cronbach’s α), and no ethics approval or informed consent.
- QSR N6 is legacy software, since replaced by NVivo.

8.4 Timeline feasibility

- Data collection for all three phases, intervention included, fits in Sep–Nov 2025 (3 months); the intervention and post-assessment are not scheduled separately.
- Tool development (Aug–Oct) overlaps data collection (Sep–Nov) by two months, leaving no pilot window.
- No slot for ethics clearance or module design; total span is 17 months.

8.5 Citation audit

Cited in text, missing from list	Listed but not cited in text
• ICMI (2016) • National Education Commission (2018) • NIE (2019) — list has NIE (2009) • DOE (2018) — list has DOE (2019) • Vygotsky (1978) • Lesh et al. (2019) • Theory of Planned Behavior / Self-Efficacy Theory (no source)	Astriani et al. 2017; Chinnapan 1998; Creswell 2009; Dissanayake & Sonnadara 2011; Desi et al. 2019; NCTM 2000; NIE 2009; Ekanayake 2014; Kalamani 2009; Kamala Liyanage 2014; Klegeris & Hurren 2011; MOE 2018; Özerem 2012; Rosli et al. 2013; Setiawan & Supiandi 2018; Sigurdson et al. 1994; Simamora & Saragih 2019; Panaoura & Gagatsis 2009; Voskoglou 2011

- List is not alphabetical (Desi after DOE; NIE before Ekanayake; Panaoura after Simamora).
- Six entries lack a journal or publisher (Astriani, Ekanayake, Kalamani, Kamala Liyanage, Klegeris, Özerem).
- Panaoura & Gagatsis: the DOI is printed as <https://doi.org/https://doi.org/...>, and 10.12973/iejme/3966 points to IEJME, not CERME 6. Needs checking.
- Creswell (2009): the 3rd edition is subtitled *Qualitative, Quantitative, and Mixed Methods Approaches* (Sage, Thousand Oaks). Needs checking.
- Four entries are about student geometry (Chinnapan, Özerem, Panaoura, Ekanayake), not teacher competency.

8.6 Editorial

- “P.hd Student” → “PhD Student”; the candidate’s name is spelled *Prashanthan* in the document but *Prasanthan* in the file name.
- “Teacher’ mathematics knowledge” → “Teachers”; “attitudes towards attitudes towards” (Phase 2.4); “problems solving” (Phase 3.3); “Convenient sample” → “convenience sample”.
- The supervisor recommendation field is signed but left empty.